

Ellyess Benmoufok

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Climate and energy data scientist with a PhD in Computational Science & Data Science from Imperial College London (viva passed June 2026, conferral 1 October 2026). I corrected ERA5 reanalysis wind against observed generation at 1,729 control points across 12 European countries, then measured what that correction changes inside European energy system models. I released the method as PyVWF, an open-source Python library with 469 tests, continuous integration, static typing, documentation and a Zenodo DOI. Three peer-reviewed publications, one as first author. Looking for data science, modelling or research software roles in climate, energy, weather and climate risk.

CORE COMPETENCIES

- **Python & Scientific Computing:** Python (advanced, since 2017), NumPy, SciPy, Pandas, Xarray, Dask, NetCDF, Zarr, Matplotlib, Jupyter. Analytical SQL (DuckDB).
- **Weather & Climate Data:** ERA5 and MERRA-2 reanalysis, CMIP6, statistical downscaling, bias correction, skill metrics and error decomposition (RMSE, MAE, bias), ENTSO-E generation data, Copernicus CDS.
- **Modelling & Validation:** scikit-learn, PyTorch, TensorFlow/Keras, GANs (WGAN-GP), physics-informed models, uncertainty quantification, spatial and leave-one-group-out cross-validation, pre-registered evaluation.
- **Geospatial:** GeoPandas, Rasterio, Cartopy, Shapely, pyproj, kriging and IDW interpolation, equal-area projections and CRS, Google Earth Engine, Landsat and Sentinel-2 classification.
- **Energy Systems & Risk:** PyPSA-Eur, Atlite, Gurobi, wake modelling, sector coupling, system-cost sensitivity analysis. Oasis LMF (PiWind reproduction): AAL, OEP and AEP.
- **Software Engineering:** pytest, GitHub Actions CI, ruff, mypy, PEP 561 typing, Sphinx, Docker, pre-commit, pyproject packaging, semantic versioning, Snakemake, Conda, Git, Unix shell. AI-assisted development with Claude Code.

EXPERIENCE

PhD Researcher, Computational Science & Data Science

Oct 2021 – Jun 2026

Imperial College London

London, UK

- Designed and released [PyVWF](#), an open-source Python library for granular spatial and temporal bias correction of reanalysis wind. Wrote all 310 commits in v0.5.1, released with a Zenodo DOI, 469 test functions, CI with ruff and mypy, and Sphinx docs built with warnings as errors.
- Trained and validated corrections at 1,729 control points across 12 European countries, gridded over 23,989 ERA5 cells, reducing capacity-factor MAE by 16 to 79% against uncorrected ERA5.
- Extended [PyPSA-Eur](#) in a public fork: a density-tiered wake-loss correction, integration of the gridded bias corrections, and a spatial splitting algorithm decoupling offshore resource resolution from network topology.
- Validated against ENTSO-E observed generation for 2023: accuracy improved in 9 of 11 country and technology combinations, with NRMSE down by up to 36%.
- Tested terrain-informed machine learning against geometric interpolation. Under spatial cross-validation the best model scored R^2 0.295, down from 0.35 to 0.41 under random folds. I reported the lower figures and kept interpolation as the published method.
- Built reproducible pipelines (Xarray, Dask, Snakemake) over multi-decade continental datasets, and seven Python API clients with rate-limit backoff, request batching and resumable downloads.
- Co-supervised an MSc research project, named alongside Prof. Matthew Piggott, that became a co-authored paper in *Energy*. Reported throughout to an industry supervisor at Shell.

Data Science Intern

Jun 2021 – Sep 2021

Shell

London, UK

- Proof of concept for statistically downscaling CMIP6 climate projections to asset level for renewable-energy variables, applying machine learning to large, heterogeneous climate datasets, with offshore wind lidar observations as ground truth.
- Supported separate operational weather-modelling work, retrieving data from HPC. Project details under NDA.

Research App Developer

Imperial College London

Dec 2020 – Mar 2021

London, UK

- Credited App Developer on [I-Hex](#), [the Island Health Explorer](#), a Google Earth Engine suite giving policymakers and island communities free access to satellite and climate data for the Maldives.
- Worked on supervised land-cover and benthic-feature classification over Maldivian atolls, and digitised and assembled its 861-point labelled training set across nine classes.

Data Scientist

Link Humans

Jan 2019 – Aug 2019

London, UK

- Most technical member of a five-person team. Built Python scrapers and pipelines to collect employer-branding data, and classified large volumes of unstructured text.
- Produced insight reports for more than ten clients on monthly to quarterly cycles, using standardised Excel reporting templates.

EDUCATION

PhD in Computational Science & Data Science

Imperial College London

Oct 2021 – Jun 2026

London, UK

- Viva passed June 2026, conferral 1 October 2026. Thesis: *Data Science-Enhanced Wind Power Modelling: From Reanalysis Correction to Energy System Representation*. Funded by an EPSRC and Shell Industrial CASE award.

MSc, Applied Computational Science and Engineering (Distinction), DIC

Imperial College London

Oct 2019 – Oct 2020

London, UK

- Modern Programming Methods 92.4, Inversion and Optimisation 82.2. Dissertation on Wasserstein GANs for multi-output regression, benchmarked against Gaussian process regression, leading to a co-authored paper.

BSc, Physics (First Class Honours)

University of Surrey

Sep 2015 – Jun 2018

Guildford, UK

SELECTED PUBLICATIONS

- **Benmoufok, E.F.**, Warder, S.C., Zhu, E., Bhaskaran, B., Staffell, I., Piggott, M.D. (2024). Improving wind power modelling through granular spatial and temporal bias correction of reanalysis data. *Energy*, 313, 133759.
- Wang, Y., Warder, S.C., **Benmoufok, E.F.**, Wynn, A., Buxton, O.R.H., Staffell, I., Piggott, M.D. (2026). Geographic variability in reanalysis wind speed biases: a high-resolution bias correction approach for UK wind energy. *Energy Conversion and Management*, 352, 121066.
- Phillips, T.R.F., Heaney, C.E., **Benmoufok, E.**, et al. (2022). Multi-Output Regression with Generative Adversarial Networks (MOR-GANs). *Applied Sciences*, 12(18), 9209.

SELECTED PROJECTS

- **Physics-informed wind correction** (PyVWF, post-PhD, experimental): gates and thirteen predictions fixed before each run. Beats uncorrected ERA5 in 5 of 5 held-out regions as registered; seven failed predictions are published as failures.
- **era5-windstorm-oasis**: reproduced the Oasis LMF PiWind example catastrophe model end to end and ran an exposure sensitivity study on AAL and EP curves, ground-up against insured. A reproduction, not a catastrophe model.
- **llm-persona-research**: evaluation harness testing LLM personas against two nationally representative UK surveys, with TVD, JSD and bootstrap confidence intervals.

CERTIFICATION

- Microsoft Certified: Azure Fundamentals (AZ-900), 2020.